Achintya Rai

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Education

University of California, San Diego — Sept. 2024 – June 2028
Bachelor of Science in Computer Engineering — ***GPA: 3.9/4.0***

Work Experience

Adobe — San Jose, CA
Software Engineering Intern — *June 2026 – Sept. 2026*

- Profiled **60.6M+** payment events to flag **86,387** duplicate records and a **47%** null rate.
- Detected a **26-hour** processing outage in Feb 2026 via hourly volume analysis across **1,272** payment monitoring segments, establishing the primary validation benchmark for the anomaly detector.
- Designed a time-series anomaly detector in **Python** across **10+** workflows and **30+** countries.
- Automated an MLOps pipeline with SQL workflows for weekly retraining and hourly inference.

VALT Health — Remote
Founding Software Engineer — *Aug. 2025 – Feb. 2026*

- Engineered a Flask backend in **Python** to ingest **100k+** datapoints from fitness devices, designing schemas to unify data formats across **20+** distinct sources.
- Automated environment setup and data seeding via Bash, cutting testing time by **80%**.
- Authored technical design documents and standards to onboard new engineers with less friction.

Decklar (formerly Roambee) — Sunnyvale, CA
Software Engineering Intern — *June 2024 – Aug. 2024*

- Developed real-time monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing **Python**/NumPy pipelines for a **15%** capacity boost across **1M+** weekly events.
- Implemented advanced data structures for analytics, delivering **20%** faster reporting.
- Enhanced workflow efficiency **15%** through Panels-based UI refinements and user testing.

Research

Graph Theory & Neural Architecture — San Diego, CA
Undergraduate Research Fellow (NSF REU) — *March 2026 – Present*

- Engineered a **Python** pipeline to analyze LLM embedding manifolds, implementing weight-matrix extraction utilities that reduce **500GB** checkpoints to **13GB** localized manifolds.
- Optimized architectural placement scripts and preprocessing for a **30x runtime reduction** (30 min to <1 min), enabling analysis on resource-constrained **8GB RAM** systems.
- Applied Rent's Rule and Louvain clustering to identify model backbones, raising Inter-Cluster Network (ICN) density from **0.18% to 13.25%** in Llama-2-70B.

Projects

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* — Aug. 2025 – Dec. 2025

- Built **CI/CD pipeline** with automated testing for **99.9%** deployment reliability.
- Architected **20+ Django REST APIs** for authentication, session, and transaction logic.

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* — Mar. 2026

- Implemented a transformer from scratch in Python/PyTorch at **100%** GPU utilization.

Technical Skills

Languages: Python, C++, C, TypeScript, SQL (PostgreSQL/SQLite), Java, JavaScript, Bash
Development & Testing: React, Django, FastAPI, Flask, Node.js, Asyncio, WebSockets, PyTest, JUnit, REST APIs
Tooling & Cloud: Docker, GitHub Actions (CI/CD), Git, Linux, AWS, GCP, PyTorch, TensorFlow, NumPy